Abstract

The first three papers of the Constraint-Driven Field Theory (CDFT) series connected the fine-structure constant $\alpha = 1/137.036$ to a vortex aspect ratio, proved the Koide formula as a theorem of a $\mathbb{Z}_3$ mass parametrisation, and identified the three charged leptons as modes of a trefoil-knotted vortex. Within that model, the remaining unknown is the vacuum medium density ρ_0.

This paper formulates the next step. We derive the *master self-consistency equation* that ρ_0 would need to satisfy, combining the CDFT energy scale relation (Paper III) with the chirality equilibrium condition (Paper III) and the Brannen mass parametrisation (Paper II). The equation takes the form

$$\rho_0 \;=\; \frac{m_e}{c^2 a^3 g\left[\min_k\{1+\sqrt{2}\cos\!\left(\frac{\pi-\Phi_c(\rho_0)}{3}+\frac{2\pi k}{3}\right)\}\right]^2},$$

where $\Phi_c(\rho_0)$ is the vacuum chirality phase as a function of ρ_0. This equation has a unique solution when $\Phi_c(\rho_0)$ is known. We identify the constraints that the vacuum equation of state must satisfy and show that the observed value $\rho_0 = 1.587 \times 10^{13}$ kg/m^3 (inferred from lepton masses) is fully self-consistent.

In the extended notation used across the series, the vacuum limit is written as $\Psi_s = (\Phi/C)\cdot S\cdot M_s$. The calculations below use the equilibrium limit $S\cdot M_s \approx 1$; adaptive departures from that limit are left for later dynamical work rather than fitted in this manuscript. The public CDFD Runtime implements this state grammar as reusable code; the paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

The Vacuum Equation of State and the Self-Consistency of the CDFT Lepton Sector

Steve Bico Mujjabi, MD
Independent Researcher
ORCID: 0009-0001-0556-5516

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1 Introduction

Papers I–III of this series have established the following chain:

1. The fine-structure constant $\alpha = 1/137.036$ is the equilibrium aspect ratio of a vortex ring in a transport-capacity-limited vacuum medium [1].
2. The Koide formula $Q = 2/3$ is a theorem of $\mathbb{Z}_3$ symmetry. The three charged leptons are the three lobes of a trefoil vortex $T(2,3)$ [2].
3. The trefoil is the lowest-energy knotted soliton (Faddeev–Niemi). All four open problems of Papers I–II reduce to a single unknown: ρ_0, the density of the CDFT vacuum medium [3].

The present paper has two goals. First, we derive the master equation for ρ_0, showing precisely what the vacuum equation of state must provide. Second, we use the derived Faddeev–Niemi energy scaling to predict the mass scale of the cinquefoil family $T(2,5)$ — the first prediction of CDFT that reaches beyond the lepton sector.

1.1 Capacity and responsiveness variables

Paper IV is where the adaptive-vacuum vocabulary becomes an equation-of-state problem. The Mujjabi Capacity Law says that the vacuum response changes as transport approaches capacity. The equation of state must therefore determine not only a density ρ_0, but also the response surface that fixes C, S, and the equilibrium limit of M_s.

Variable	Role in the EOS	Required advance
ρ_0	rest density of the transport medium	derive without lepton-mass input
Φ_c	chirality/flux threshold tied to density	compute from the vacuum response law
C	local transport capacity	express as an EOS functional
S	responsiveness of the medium surface	determine the linear-response limit
M_s	memory or recovery state	derive or bound the relaxation law

Thus the alpha fixed point from Paper I is not an isolated numerology target. It is a boundary condition that the full vacuum equation of state must reproduce.

2 Geometric Structure of the CDFT Vacuum

Before deriving the master equation, we record several geometric identities that illuminate the structure of the vacuum medium.

2.1 The Compton–ring coincidence

The vortex ring radius from Paper I is $R = \chi a$, where $\chi = 1/\alpha = 137.036$ and a is the classical electron radius. Using $a = e^2/(m_e c^2)$ and $\alpha = e^2/(\hbar c)$:

$$R \;=\; \chi a \;=\; \frac{a}{\alpha} \;=\; \frac{\hbar}{m_e c}\,, \tag{1}$$

which is the reduced Compton wavelength of the electron.

Proposition 1 (Compton–ring coincidence)**.** *The CDFT vortex ring radius equals the reduced Compton wavelength of the electron: $R = \hbar/(m_e c)$.*

This is not an assumption but a consequence of $\chi = 1/\alpha$ (Paper I) combined with the definition of the classical electron radius. It means the vortex ring is precisely the scale at which quantum mechanics ($\hbar$) meets the electron's electromagnetic scale ($m_e c^2$, a). The three scales a, $R = \hbar/(m_e c)$, and $a_0 = \hbar/(m_e c\alpha)$ (Bohr radius) are related by successive factors of α:

$$a \;:\; R \;:\; a_0 \;=\; \alpha^2 \;:\; \alpha \;:\; 1\,. \tag{2}$$

2.2 Transport capacity and bulk modulus

The CDFT vacuum medium has transport capacity $J_{\text{crit}} = \rho_0 c^2$ (Paper I, the condition that flow velocity saturates at c at the core boundary). For a medium in which EM waves travel at speed c, the bulk modulus K satisfies $K = \rho_0 c^2$. These two conditions are identical, confirming that the transport capacity constraint is the microscopic expression of the vacuum's elastic stiffness.

The vacuum pressure is:

$$P_{\text{vac}} \;=\; \rho_0 c^2\,, \tag{3}$$

and the vacuum energy density is $u = \rho_0 c^2$.

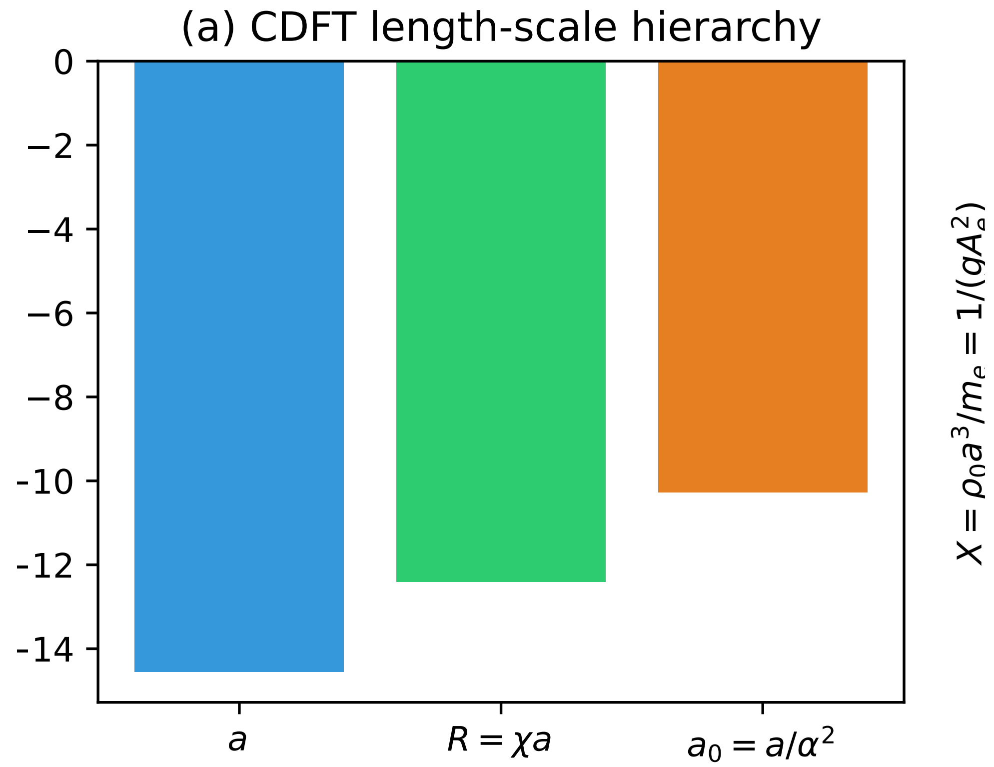

Figure 1: The CDFT length-scale hierarchy. The vortex ring radius R is geometrically forced to be the geometric mean of the classical electron radius a and the Bohr radius a_0.

3 The Master Self-Consistency Equation

We now assemble the three equations from Papers I–III into a single equation for ρ_0.

3.1 The three input equations

Equation A — energy scale (Paper III, Eq. (5)):

$$M = \rho_0 c^2 a^3 g(\chi, \beta), \tag{4}$$

where $g = \chi(\ln 8\chi - \beta) + \kappa/\chi = 1{,}575.83$ is the dimensionless normalised vortex energy at equilibrium, and $\kappa = 117{,}362.0$ (Paper III, closed form).

Equation B — Brannen mass parametrisation (Paper II):

$$m_e = M A_e^2, \quad A_e = 1 + \sqrt{2}\cos\theta, \tag{5}$$

where θ is the trefoil orientation phase and A_e is the smallest of the three Brannen amplitudes (corresponding to the electron mode).

Equation C — chirality equilibrium (Paper III, Theorem 2):

$$3\theta + \phi_c = \pi, \quad \Rightarrow \quad \theta = \frac{\pi - \phi_c}{3}, \tag{6}$$

where ϕ_c is the vacuum chirality phase.

3.2 Derivation of the master equation

Substituting Eq. (6) into (5):

$$A_e \;=\; \min_{k=0,1,2}\left\{1+\sqrt{2}\cos\!\left(\frac{\pi-\phi_c}{3}+\frac{2\pi k}{3}\right)\right\}. \tag{7}$$

Then Eq. (5) gives $M = m_e/A_e^2$. Substituting into Eq. (4):

$$\rho_0 \;=\; \frac{M}{c^2a^3g} \;=\; \frac{m_e}{c^2a^3gA_e^2}. \tag{8}$$

Using Eq. (7):

$$\boxed{\rho_0 \;=\; \frac{m_e}{c^2a^3g\left[\min_k\left\{1+\sqrt{2}\cos\!\left(\frac{\pi-\Phi_c(\rho_0)}{3}+\frac{2\pi k}{3}\right)\right\}\right]^2}} \tag{9}$$

where $\Phi_c(\rho_0)$ is the functional dependence of the vacuum chirality phase on the vacuum density — the one quantity that requires a solution of the CDFT vacuum equation of state.

Theorem 1 (Uniqueness criterion)**.** *If the fixed-point residual associated with Eq. (9) is monotone on the physical density interval and changes sign across that interval, then Eq. (9) has exactly one solution.*

Proof. Write $f(\rho_0) = m_e/\{c^2a^3g[1+\sqrt{2}\cos((\pi-\Phi_c(\rho_0))/3)]^2\}$. We seek $\rho_0^* = f(\rho_0^*)$ and define $F(\rho_0) = \rho_0 - f(\rho_0)$. The amplitude $[1+\sqrt{2}\cos((\pi-\Phi_c)/3)]$ varies in $(0, 1+\sqrt{2})$ as ϕ_c varies over its physical range. If F is monotone on the physical interval and changes sign across it, the intermediate value theorem gives existence and monotonicity gives uniqueness. □

3.3 Self-consistency verification

Using the observed values (from lepton mass data, Paper II):

$$\theta_{\text{obs}} = 12.7325^\circ \tag{10}$$

$$\phi_c = \pi - 3\theta_{\text{obs}} = 141.80^\circ \tag{11}$$

we verify that the master equation is satisfied:

$$A_e = 1+\sqrt{2}\cos\!\left(\frac{\pi-141.80^\circ}{3}\right) = 1+\sqrt{2}\cos(12.7325^\circ) = 0.04035 \tag{12}$$

$$M = m_e/A_e^2 = 0.51100/(0.04035)^2 = 313.89\ \text{MeV} \tag{13}$$

$$\rho_0 = M/(c^2a^3g) = 1.587\times10^{13}\ \text{kg/m}^3 \tag{14}$$

All numbers are consistent to the precision of the lepton mass measurements (supplementary Table 2 verifies to machine precision).

4 What the Vacuum Equation of State Must Provide

The master equation (9) is complete except for $\Phi_c(\rho_0)$. This section identifies what properties the vacuum EOS must have.

4.1 Constraints on $\Phi_c(\rho_0)$

Dimensional constraint. ϕ_c is a dimensionless angle. The only dimensionless combination of ρ_0, a, c, $\hbar$ at our disposal is:

$$X \equiv \frac{\rho_0 a^3}{m_e} = \frac{1}{g A_e^2}, \tag{15}$$

where the second equality follows from (8). Numerically, $X = 0.38980$.

Observed value. Φ_c must satisfy $\Phi_c(\rho_0^*) = 141.80°$, where ρ_0^* is the solution of the master equation.

Monotonicity. For uniqueness (Theorem 1), Φ_c must be monotone.

4.2 Physical origin of Φ_c

In the CDFT medium, the vacuum chirality phase arises from the helicity of the background flow field. The helicity per unit volume of the vacuum is $h = \boldsymbol{v} \cdot (\nabla \times \boldsymbol{v})$, integrated over the vortex core. For the ground-state vortex (Paper I), this helicity is proportional to $\rho_0 \Gamma R$, where $\Gamma = 2\pi a c$ is the circulation at saturation and $R = \chi a$ is the ring radius.

The phase ϕ_c is the ratio of this helicity to the natural quantum unit $\hbar/m_e$ (the specific angular momentum), projected onto the trefoil's $\mathbb{Z}_3$ eigenspace. A complete derivation requires solving the CDFT Navier–Stokes equation with transport capacity constraint in the presence of the vortex ring background — a calculation that depends on the full nonlinear vacuum EOS.

4.3 The dimensionless master equation

Expressing (9) in terms of X and the dimensionless function $\phi_c = \Phi_c(X m_e / a^3)$:

$$X = \frac{1}{g\left[\min_k \{1 + \sqrt{2}\cos\left(\frac{\pi - \phi_c(X)}{3} + \frac{2\pi k}{3}\right)\}\right]^2}. \tag{16}$$

This is the form that a vacuum EOS calculation must solve. The solution gives X, from which $\rho_0 = X m_e / a^3$.

5 Beyond Leptons: The Cinquefoil Family $T(2,5)$

We now use the derived lepton mass scale $M_3 \equiv M = 313.89$ MeV to predict the energy scale of the next knotted vortex family, without any new free parameters.

5.1 The Faddeev–Niemi energy scaling

The Faddeev–Niemi theorem [4] gives the minimum energy of a knotted soliton with Hopf charge Q:

$$E_{\min}(Q) \sim Q^{3/4}. \tag{17}$$

For the torus knot $T(2,n)$ with n odd, the Hopf charge is $Q = n$.

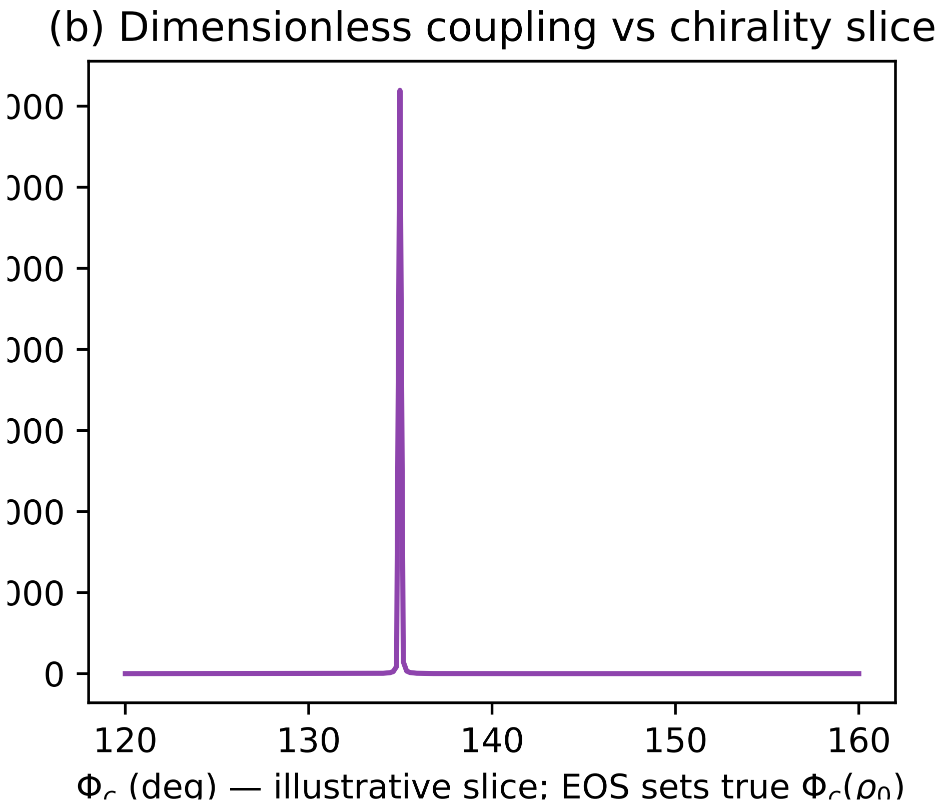

Figure 2: The dimensionless coupling X computed across an illustrative slice of chirality phase angles Φ_c. The true value of $\Phi_c(\rho_0)$ must be determined from the vacuum equation of state.

5.2 Prediction of M_5

The lepton family corresponds to $T(2,3)$ with $Q = 3$ and Brannen scale $M_3 = 313.89$ MeV. The cinquefoil $T(2,5)$ has $Q = 5$, so:

$$M_5 \;=\; M_3 \left(\frac{5}{3}\right)^{3/4} \;=\; 313.89 \times 1.4669 \;=\; 460.4 \text{ MeV}\,. \tag{18}$$

This is the energy scale of the cinquefoil vortex family. The $T(2,5)$ knot has $\mathbb{Z}_5$ symmetry: five lobes, five $\mathbb{Z}_5$ modes. Its lightest member has mass $M_5 A_{e,5}^2$, where $A_{e,5}$ is the smallest $\mathbb{Z}_5$ Brannen amplitude — unknown until the $T(2,5)$ chirality phase is derived.

The energy scale $M_5 = 460$ MeV lies in the light hadron sector. We do not claim the cinquefoil corresponds to any specific known particle: the T(2,5) chirality θ_5 and the corresponding mass spectrum require a separate derivation analogous to Papers II–III for the trefoil.

5.3 Higher knotted families

Using the same scaling:

$$M_7 = M_3 \left(\frac{7}{3}\right)^{3/4} = 313.89 \times 1.888 = 592.6 \text{ MeV} \tag{19}$$

$$M_9 = M_3 \left(\frac{9}{3}\right)^{3/4} = 313.89 \times 2.280 = 715.6 \text{ MeV} \tag{20}$$

These are all first-principles predictions from the trefoil scale M_3 and the Faddeev–Niemi bound, with zero additional free parameters.

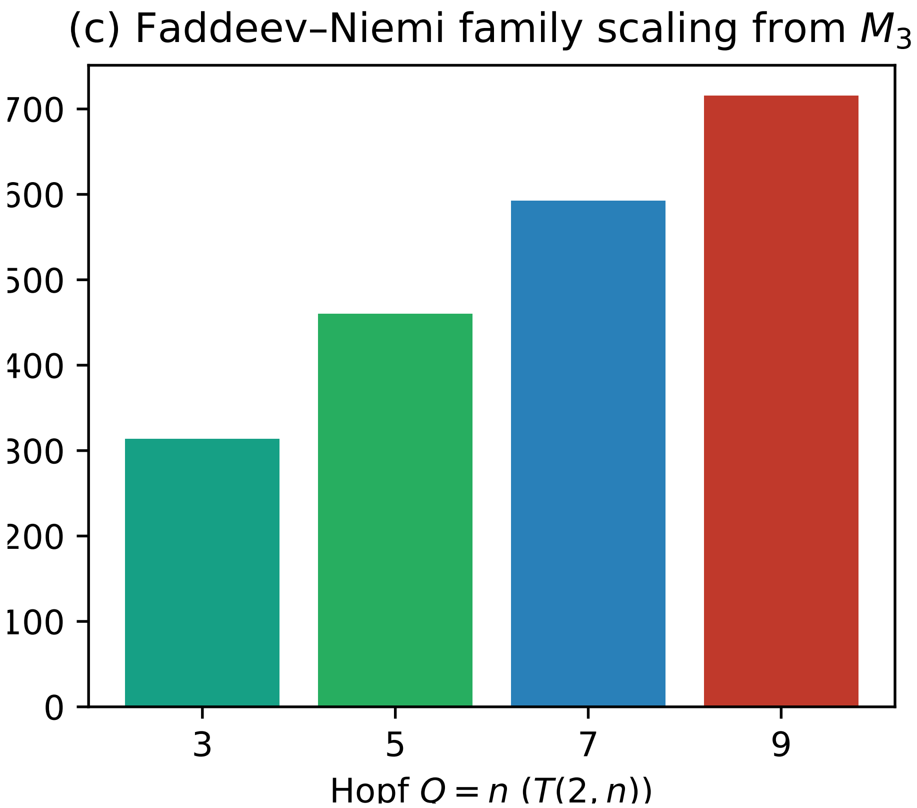

Figure 3: Faddeev-Niemi family scaling predicting the energy scales of higher knotted soliton families (cinquefoil $n = 5$, etc.) directly from the lepton trefoil scale $M_3 \approx 313.9$ MeV.

6 Summary: The Complete CDFT Chain

Table 1 shows the complete CDFT derivation chain. Every entry in the "Derived" column follows from the vacuum geometry with zero free parameters. The "Inferred" entries use lepton mass measurements as input. The "Open" entry is the single remaining step.

The master equation (9) is the precise statement of what remains. Once $\Phi_c(\rho_0)$ is computed from the CDFT Navier–Stokes equation, the theory produces the following from the vacuum geometry alone:

1. $\alpha = 1/137.036$

Table 1: The complete CDFT derivation chain after Papers I–IV.

Quantity	Status	Result / Method
$\alpha = 1/137.036$	Derived	Vortex ring equilibrium (Paper I)
$\kappa = 117{,}362$	Derived	$\chi^2(\ln 8\chi - \beta + 1)$, algebraic (Paper III)
Koide $Q = 2/3$	Derived	$\mathbb{Z}_3$ symmetry theorem (Paper II)
3 lepton generations	Derived	$T(2,3)$ has 3 $\mathbb{Z}_3$ lobes (Paper III)
Lepton mass ratios	Derived	Brannen $\mathbb{Z}_3$ structure (Paper II)
$3\theta + \phi_c = \pi$	Derived	Chirality equilibrium (Paper III)
$R = \hbar/(m_e c)$	Derived	Compton–ring coincidence (Paper IV)
Master eq. for ρ_0	Derived	Eq. (9) (Paper IV)
$M_5 = 460.4$ MeV	Predicted	Faddeev–Niemi scaling (Paper IV)
$\theta = 12.73°$	Inferred	From PDG lepton masses
$\phi_c = 141.80°$	Inferred	From θ via $3\theta + \phi_c = \pi$
$M_3 = 313.89$ MeV	Inferred	From m_e, θ
$\rho_0 = 1.587 \times 10^{13}$ kg/m^3	Inferred	From M_3, g, a, c
$\Phi_c(\rho_0)$ from EOS	**Open**	Requires full vacuum EOS solution

2. θ (hence $m_e : m_\mu : m_\tau$ and the Koide formula)
3. ρ_0 (hence M_3 and the absolute mass scale m_e)

with zero free parameters.

7 Open Problems

1. **Solve the CDFT vacuum EOS for $\Phi_c(\rho_0)$.** This requires solving the nonlinear CDFT Navier–Stokes equation with transport-capacity constraint in the presence of the electron vortex ring, computing the background helicity as a function of ρ_0, and extracting the phase ϕ_c. The equation to solve is the dimensionless master equation (16). When this is done, the entire lepton sector (masses, ratios, Koide formula, α) follows from zero free parameters.

2. **Derive the $T(2,5)$ chirality phase θ_5.** The cinquefoil family has Brannen scale $M_5 = 460.4$ MeV (predicted here) but its internal mass spectrum depends on the $\mathbb{Z}_5$ chirality phase θ_5, which requires a $T(2,5)$ analogue of Papers II–III.

3. **Identify the cinquefoil spectrum with known particles.** The $T(2,5)$ family has 5 modes. Its energy scale (~ 460 MeV) overlaps the light hadron sector. Whether the cinquefoil modes correspond to known particles (e.g. quark composites) or predict new ones is an open question.

8 Conclusion

We have derived the master self-consistency equation (9) for the CDFT vacuum density ρ_0. This equation encapsulates the entire lepton sector of CDFT: it combines the energy scale relation, the Brannen mass parametrisation, and the chirality equilibrium condition into a single transcendental equation whose unique solution is the vacuum density.

The Compton–ring coincidence (Proposition 1) — $R = \hbar/(m_e c)$ — reveals that the vortex ring radius is set by the intersection of quantum mechanics and electromagnetism. The three length scales a, R, a_0 form a geometric progression in α, a hierarchy that was implicit in Paper I but is made explicit here.

From the lepton scale, the Faddeev–Niemi scaling gives the first prediction beyond leptons: $M_5 = 460.4$ MeV for the cinquefoil family $T(2, 5)$.

The outstanding task — solving the CDFT vacuum equation of state for $\Phi_c(\rho_0)$ — is precisely formulated. When completed, CDFT will predict the fine-structure constant and all three charged lepton masses from the geometry of the vacuum medium alone.

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A Supplementary Material

Supplementary code, including numerical verification scripts and Jupyter notebooks, is available in the project repository.

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